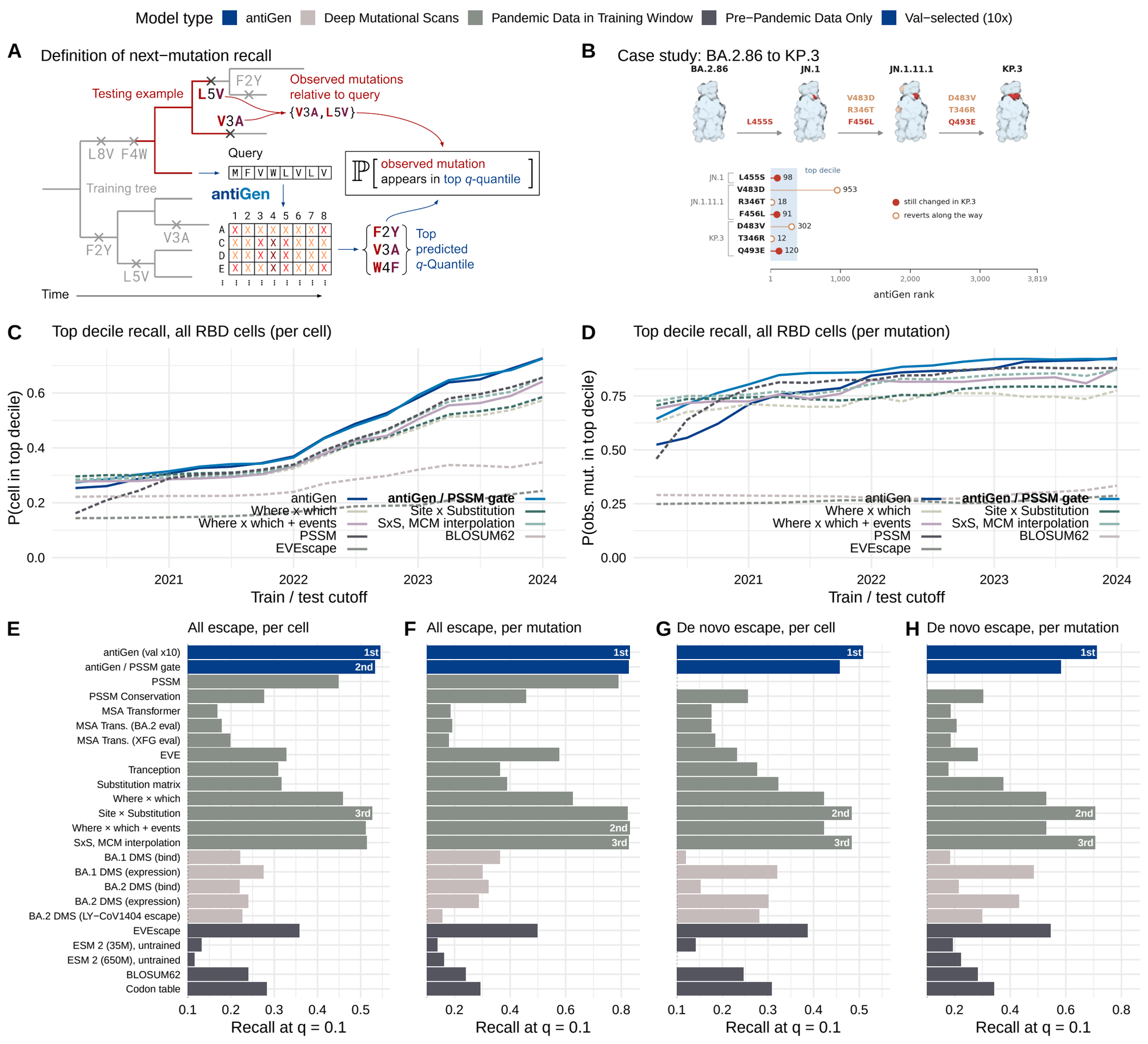

Model type
antiGen
Deep Mutational Scans
Pandemic Data in Training Window
Pre−Pandemic Data Only
Val−selected (10x)
A
Definition of next−mutation recall
Testing example
L5V
F2Y
V3A
Observed mutations relative to query
{V3A, L5V}
L8V F4W
Training tree
Query
M F V W L V L V
antiGen
observed mutation appears in top q-quantile
1 2 3 4 5 6 7 8
A
C
D
E
F2Y
V3A
W4F
Top predicted q-Quantile
Time
B
Case study: BA.2.86 to KP.3
BA.2.86
JN.1
JN.1.11.1
KP.3
L455S
V483D
R346T
F456L
D483V
T346R
Q493E
top decile
JN.1
JN.1.11.1
KP.3
still changed in KP.3
reverts along the way
1
1,000
2,000
3,000
3,819
antiGen rank
C
Top decile recall, all RBD cells (per cell)
P(cell in top decile)
0.0
0.2
0.4
0.6
2021
2022
2023
2024
Train / test cutoff
antiGen
Where x which
Where x which + events
PSSM
EVEscape
antiGen / PSSM gate
Site x Substitution
SxS, MCM interpolation
BLOSUM62
D
Top decile recall, all RBD cells (per mutation)
P(obs. mut. in top decile)
0.00
0.25
0.50
0.75
2021
2022
2023
2024
Train / test cutoff
antiGen
Where x which
Where x which + events
PSSM
EVEscape
antiGen / PSSM gate
Site x Substitution
SxS, MCM interpolation
BLOSUM62
E
All escape, per cell
F
All escape, per mutation
G
De novo escape, per cell
H
De novo escape, per mutation
antiGen (val x10)
antiGen / PSSM gate
PSSM
PSSM Conservation
MSA Transformer
MSA Trans. (BA.2 eval)
MSA Trans. (XFG eval)
EVE
Tranception
Substitution matrix
Where × which
Site × Substitution
Where × which + events
SxS, MCM interpolation
BA.1 DMS (bind)
BA.1 DMS (expression)
BA.2 DMS (bind)
BA.2 DMS (expression)
BA.2 DMS (LY−CoV1404 escape)
EVEscape
ESM 2 (35M), untrained
ESM 2 (650M), untrained
BLOSUM62
Codon table
1st
2nd
3rd
0.1
0.2
0.3
0.4
0.5
0.6
0.8
Recall at q = 0.1